MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 1

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Outline

01 Introduction and Logistics

02 Generative Models Overview

03 Generative Modeling Framework

Course Tricks
Problem Statement
Divergence Minimization Framework

04 Autoregressive Models (ImageGPT)

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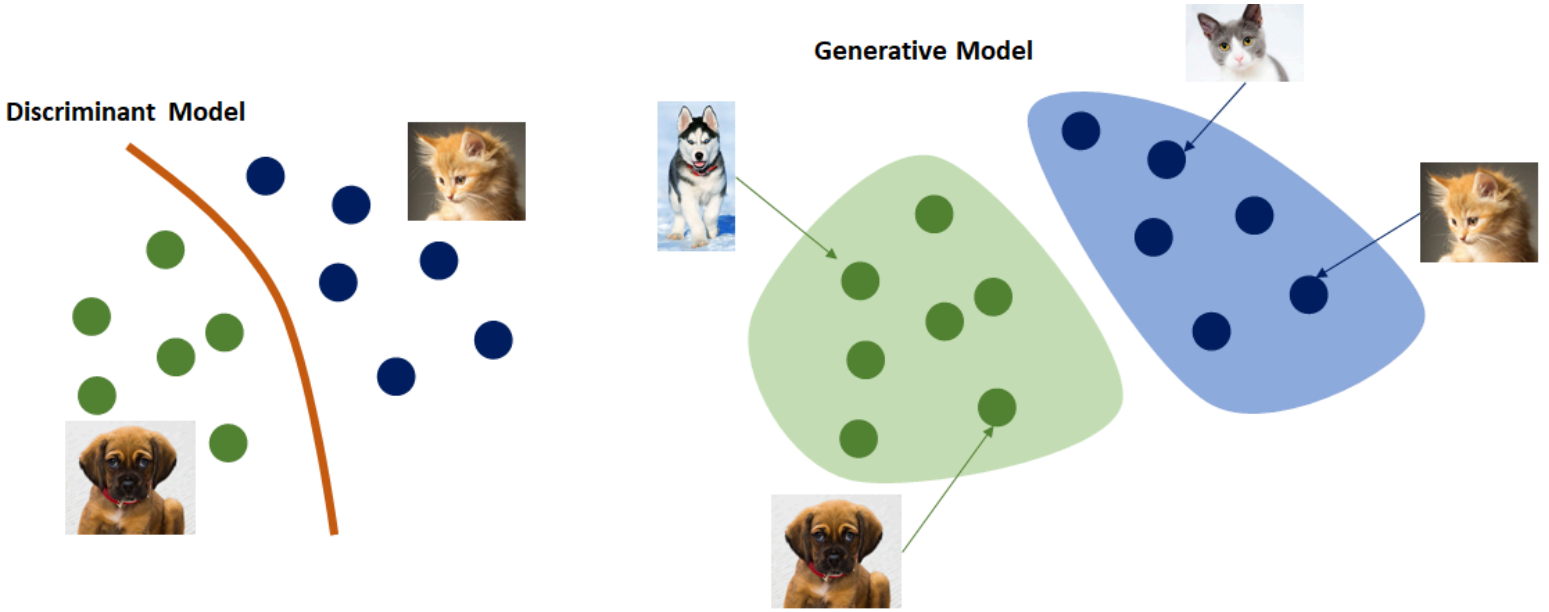
Generative vs Discriminative Models



- **Discriminative model:** learns a decision boundary $p(\mathbf{y}|\mathbf{x})$.
- **Generative model:** learns the data distribution $p_{\text{data}}(\mathbf{x})$ (optionally $p_{\text{data}}(\mathbf{x}, \mathbf{y})$) and supports **sampling (generation)**.

Course Team

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About the Course

Theoretical foundations & practical applications of **Deep Generative Models**. Primary focus: Computer Vision, but the principles are universal across modern AI.

Six major pillars

- **Likelihood-based:** Autoregressive, Normalizing Flows.
- **Latent Variable:** VAE, VQ-VAE.
- **Adversarial:** GAN, WGAN, model quality evaluation.
- **Score-based & Diffusion:** score matching, NCSN, DDPM, guidance.
- **Continuous-Time:** CNF, SDE/ODE, Flow Matching.
- **Discrete Diffusion:** absorbing, masked diffusion LMs.

Course Structure and Grading

Structure

- **14** lectures + **14** seminars (one per week)
- **6** homework assignments
- cozy oral exam

Game rules

	Points
Homeworks (6×15 pts)	**90**
Oral exam	**30**
Maximum	**120**

Final grade

$$\text{grade} = \min\left(\left\lfloor \frac{\#\text{points}}{10} \right\rfloor, 10\right)$$

Lecture Schedule

1. Intro & Logistics; AR models (ImageGPT)
2. Normalizing Flows (NF); LVM intro
3. ELBO, Reparametrization Trick, VAE
4. ELBO Surgery; Discrete VAE; VQ-VAE; Likelihood-free Learning
5. GAN; Wasserstein Distance; WGAN; Model Quality Evaluation
6. Langevin Dynamics; Score Matching; NCSN
7. Gaussian Diffusion; Diffusion ELBO
8. DDPM; Classifier(-Free) Guidance
9. Continuous-Time NF; Continuity Equation; SDE basics
10. Probability Flow ODE; Reverse SDE; Flow Matching
11. Conditional Flow Matching (CFM)
12. Two-Sided Conditioning; FM $\leftrightarrow$ Score link; Discrete Diffusion
13. Discrete Diffusion: Reverse Process; Absorbing Diffusion
14. Course Overview and Recap

Prerequisites

What you should know

- **Math:** probability theory, statistics, multivariate calculus.
- **ML:** fundamentals of machine learning and deep learning.
- **Tools:** PyTorch / basic GPU workflow is a plus.
- **Optional:** information theory basics (entropy, KL divergence).
- **Optional:** familiarity with numerical ODE concepts at a “black-box usage” level.

Key points

- The course is **mathematically heavy**.
- The course is **constantly evolving**.
- Any feedback — especially *negative* — is very welcome!

Repository: github.com/r-isachenko/2026-DGM-MIPT-YSDA-course

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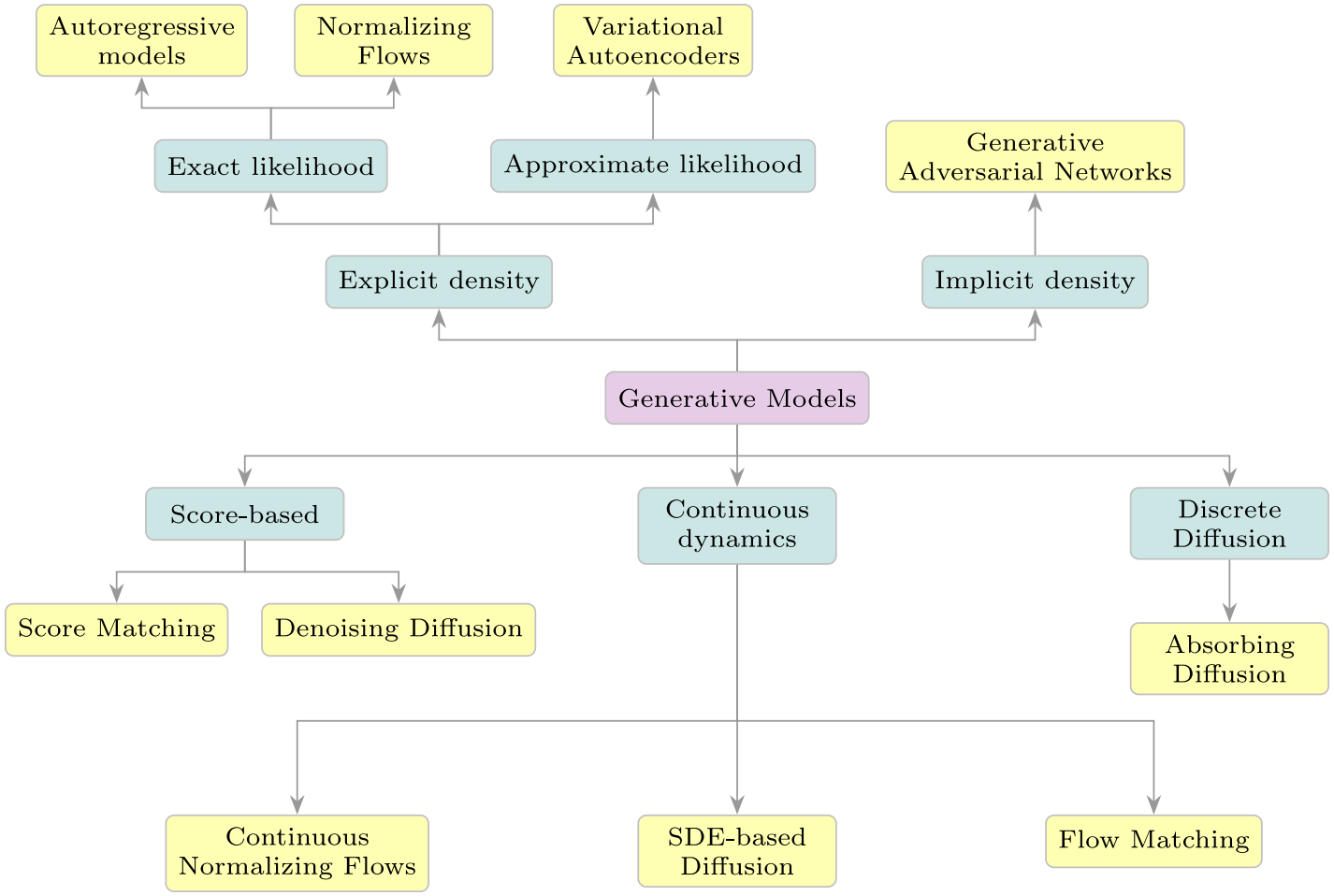
Generative Models Taxonomy



VAE — The First Scalable Approach for Image Generation

Kingma D.P., Welling M. Auto-Encoding Variational Bayes, 2013

DCGAN — The First Convolutional GAN for Image Generation

Radford A., Metz L., Chintala S. Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks, 2015

StyleGAN — High-Quality Face Generation

Language Modeling at Scale

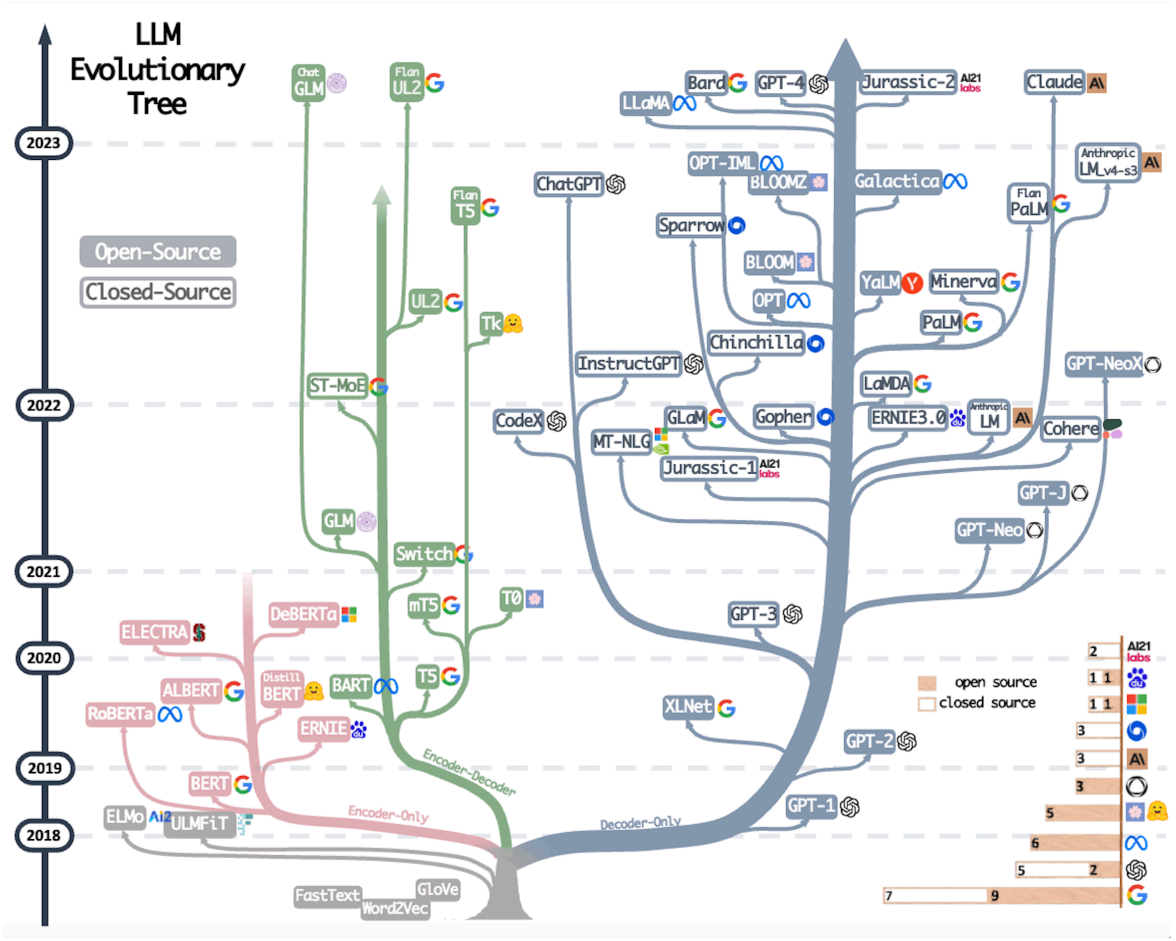


Image credit: https://blog.biocomm.ai/2023/05/14/open-source-proliferation-llm-evolutionary-tree/

Language Models: 2023–2026

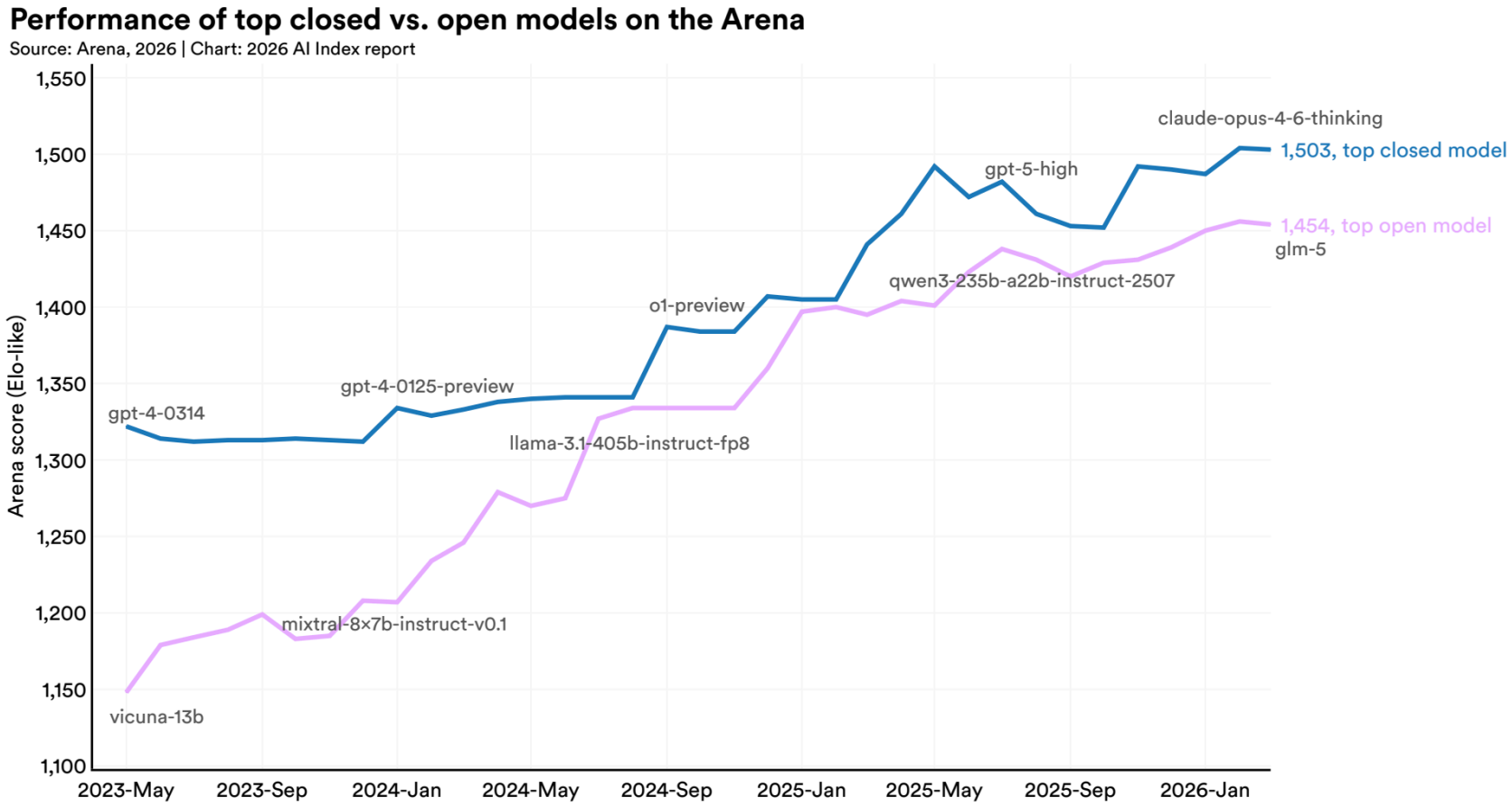


Figure 2.1.2[2]

Stanford HAI. AI Index Report 2026, Figure 2.1.2, p. 77. Source: Arena historical leaderboard, March 2026.
Arena scores summarize human preferences in pairwise comparisons.

Diffusion Models — A Breakthrough in Image Generation

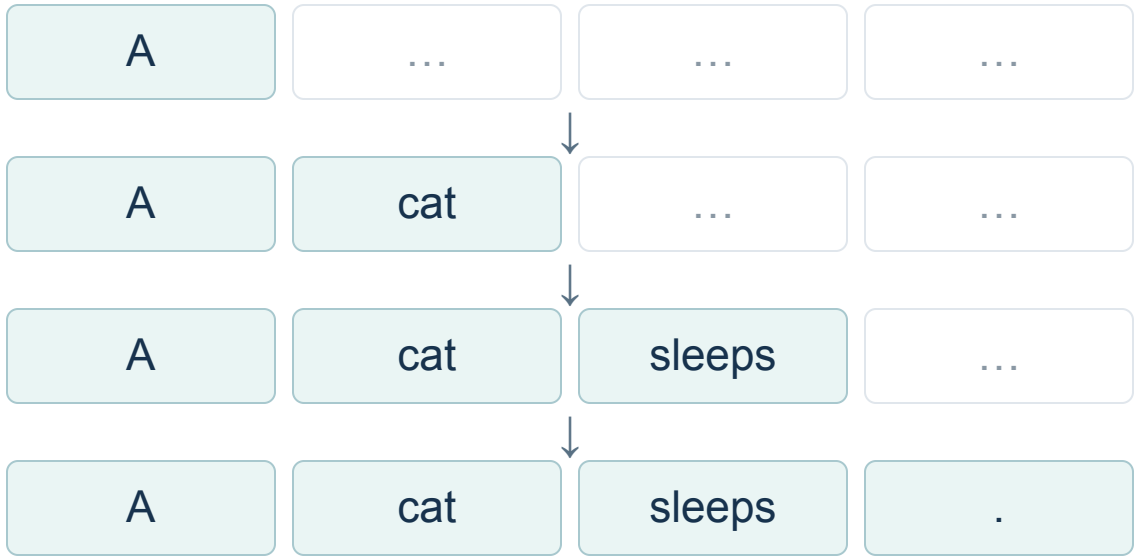
DDPM (2020) → Diffusion beats GANs (2021) → Midjourney (2022)

Ho J. et al. Denoising Diffusion Probabilistic Models, 2020 · Dhariwal P., Nichol A. Diffusion Models Beat GANs on Image Synthesis, 2021
Midjourney gallery · Midjourney model history (2022)

Diffusion Language Models — LLaDA (2025)

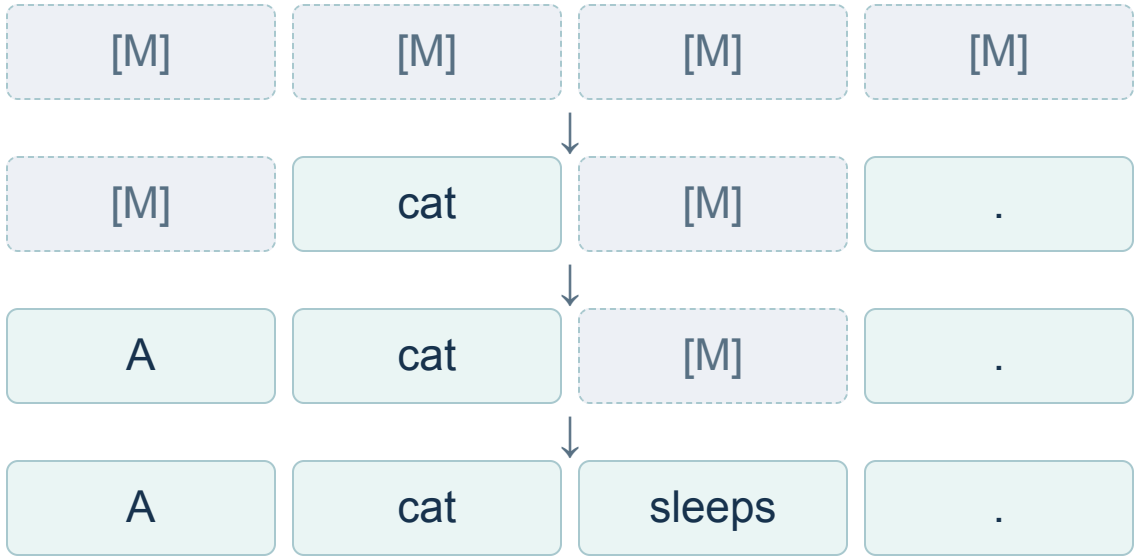
Language generation can use **iterative unmasking**.

Autoregressive



Append tokens from left to right.

Masked diffusion



Restore tokens across the sequence.

Nie S. et al. Large Language Diffusion Models, 2025. · LLaDA project page
Schematic selected sampling steps; [M] denotes a masked token. Example words stand in for tokens.

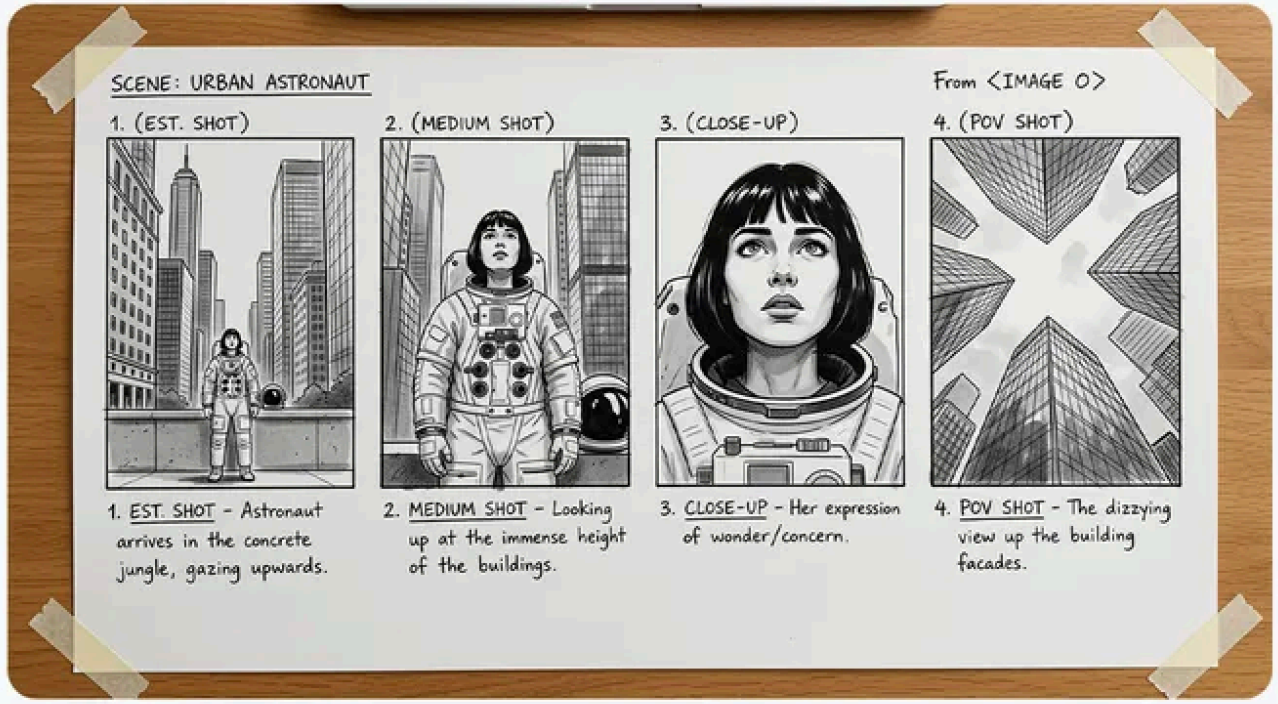
Sora — Video Generation (2024)

Image credit: https://openai.com/index/sora

Nano Banana Pro (2025)

Prompt: Create a storyboard for this scene

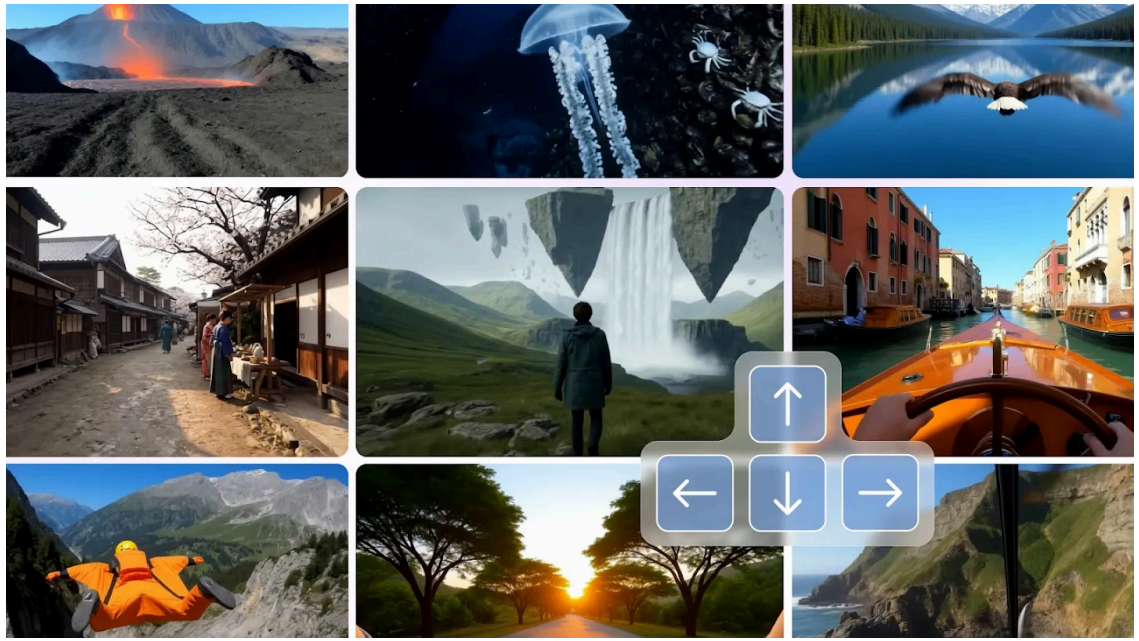
Input image



Output image

Image credit: https://blog.google/technology/ai/nano-banana-pro/

Genie 3 — Interactive World Generation (2025)

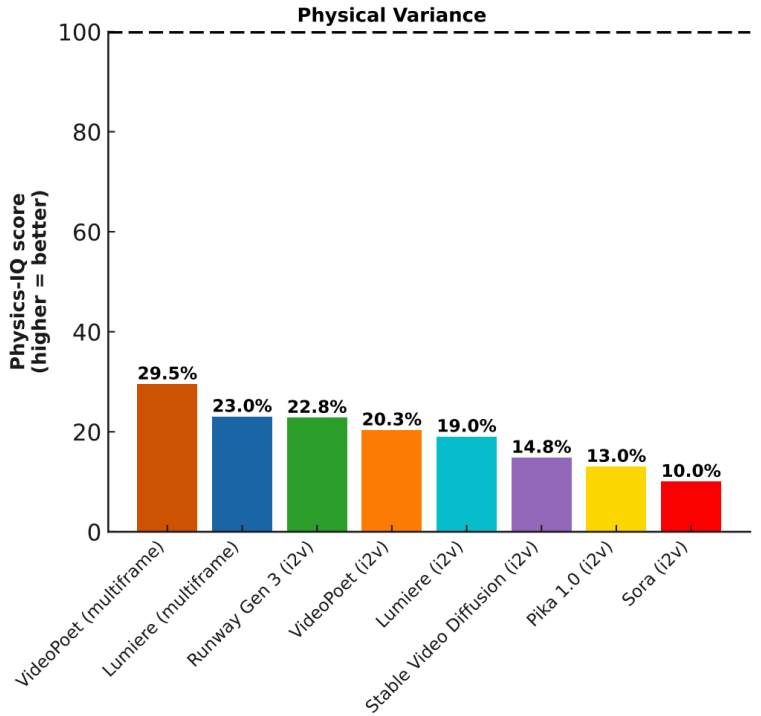


- Start from a text prompt.
- Navigate the generated environment.
- Future frames respond to your actions.

World model: predicts how observations evolve under actions.

Open Problems in Generative Models

- Video generation
- 3D scene generation
- Understanding of physical processes
- Multimodal end-to-end models



Motamed S. et al. Do generative video models understand physical principles?, 2025

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Course Tricks I

We omit density subscripts: $p(\mathbf{x}) \equiv p_{\mathbf{x}}(\mathbf{x})$.

Log-Derivative Trick (LGT)

Given a differentiable function $p : \mathbb{R}^m \to \mathbb{R}$ (usually density function),

$$\nabla \log p(\mathbf{x}) = \frac{1}{p(\mathbf{x})} \cdot \nabla p(\mathbf{x}).$$

Jensen's Inequality (JI)

If $\mathbf{x} \in \mathbb{R}^m$ is a continuous random variable with density $p(\mathbf{x})$ and $f : \mathbb{R}^m \to \mathbb{R}$ is convex, then

$$\mathbb{E}[f(\mathbf{x})] \geq f(\mathbb{E}[\mathbf{x}]).$$

Monte Carlo Estimation (MC)

Let $\mathbf{x} \in \mathbb{R}^m$ be a continuous random variable with density $p(\mathbf{x})$, and $\mathbf{f} : \mathbb{R}^m \to \mathbb{R}^d$ be any vector-valued function. Then,

$$\mathbb{E}_{p(\mathbf{x})}\mathbf{f}(\mathbf{x}) = \int p(\mathbf{x})\mathbf{f}(\mathbf{x})d\mathbf{x} \approx \frac{1}{n}\sum_{i=1}^{n}\mathbf{f}(\mathbf{x}_i), \quad \text{where } \mathbf{x}_i \sim p(\mathbf{x}).$$

Course Tricks II

Law of the Unconscious Statistician (LOTUS)

Let $\mathbf{x} \sim p(\mathbf{x})$, $\mathbf{y} = \mathbf{f}(\mathbf{x})$, and $\mathbf{f}, \mathbf{g}$ be measurable with $\mathbb{E}\|\mathbf{g}(\mathbf{y})\| < \infty$. Then

$$\mathbb{E}[\mathbf{g}(\mathbf{y})] = \mathbb{E}_{p(\mathbf{x})}\mathbf{g}(\mathbf{f}(\mathbf{x})) = \int p(\mathbf{x})\mathbf{g}(\mathbf{f}(\mathbf{x}))d\mathbf{x}.$$

Change of Variables Theorem (CoV)

Let $\mathbf{x} \in \mathbb{R}^m$ be a random vector with density $p(\mathbf{x})$, and let $\mathbf{f} : \mathbb{R}^m \to \mathbb{R}^m$ be a C^1-diffeomorphism ($\mathbf{f}$ and $\mathbf{f}^{-1}$ are continuously differentiable mappings). If $\mathbf{z} = \mathbf{f}(\mathbf{x})$, then

$$p(\mathbf{x}) = p(\mathbf{z})|\det(\mathbf{J}_{\mathbf{f}})| = p(\mathbf{z})\left|\det\left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}}\right)\right| = p(\mathbf{f}(\mathbf{x}))\left|\det\left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}}\right)\right|$$

$$p(\mathbf{z}) = p(\mathbf{x})|\det(\mathbf{J}_{\mathbf{f}^{-1}})| = p(\mathbf{x})\left|\det\left(\frac{\partial \mathbf{x}}{\partial \mathbf{z}}\right)\right| = p(\mathbf{f}^{-1}(\mathbf{z}))\left|\det\left(\frac{\partial \mathbf{f}^{-1}(\mathbf{z})}{\partial \mathbf{z}}\right)\right|$$

Course Tricks III

Optimal MSE Prediction

Let $(\mathbf{x}, \mathbf{y})$ be jointly distributed with density $p(\mathbf{x}, \mathbf{y})$. The optimal predictor under mean squared error is the conditional expectation:

$$\mathbf{f}^*(\mathbf{x}) = \arg\min_{\mathbf{f}} \mathbb{E}_{p(\mathbf{x},\mathbf{y})} \|\mathbf{y} - \mathbf{f}(\mathbf{x})\|^2 = \mathbb{E}_{p(\mathbf{y}|\mathbf{x})}[\mathbf{y}].$$

Dirac Delta Function

Any deterministic variable $\mathbf{x}_0$ can be interpreted as a random variable with density $p(\mathbf{x}) = \delta(\mathbf{x} - \mathbf{x}_0)$.

$$\delta(\mathbf{x}) = \begin{cases} +\infty, & \mathbf{x} = 0 \\ 0, & \mathbf{x} \neq 0 \end{cases} \qquad \int \delta(\mathbf{x}) d\mathbf{x} = 1$$

$$\mathbb{E}_{p(\mathbf{x})} \mathbf{f}(\mathbf{x}) = \int \delta(\mathbf{x} - \mathbf{x}_0) \mathbf{f}(\mathbf{x}) d\mathbf{x} = \mathbf{f}(\mathbf{x}_0)$$

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Problem Statement

We're given a **finite** number of i.i.d. samples $\{\mathbf{x}_i\}_{i=1}^n \subset \mathbb{R}^m$ drawn from an **unknown** distribution $p_{\text{data}}(\mathbf{x})$.

Objective

Learn a model $p(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$ from data (**density estimation**). Two model operations:

- **Sampling (generation):** sample a new object $\mathbf{x} \sim p(\mathbf{x})$.
- **Density evaluation:** compute $p(\mathbf{x})$ for a given object $\mathbf{x}$.

Challenge

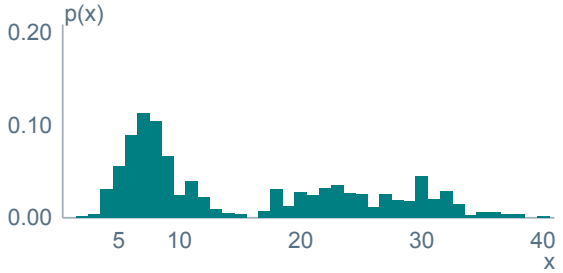
The data is high-dimensional and complex. For example, image datasets live in $\mathbb{R}^{\text{width} \times \text{height} \times \text{channels}}$. The curse of dimensionality makes accurately estimating $p_{\text{data}}(\mathbf{x})$ infeasible.

Histogram as a Generative Model

Assume $x \sim \mathrm{Cat}(\boldsymbol{\pi})$. The histogram model is fully characterized by

$$\hat{\pi}_k = \hat{\pi}(x = k) = \frac{\sum_{i=1}^{n} [x_i = k]}{n}.$$

Curse of dimensionality: The number of bins rises exponentially.



n = 1000 observations

MNIST example: 28×28 grayscale images, with each image $\mathbf{x} = (x_1, \ldots, x_{784})$, $x_i \in \{0, 1\}$:

$$p_{\text{data}}(\mathbf{x}) = p(x_1) \cdot p(x_2|x_1) \cdot \cdots \cdot p(x_m|x_{m-1}, \ldots, x_1).$$

A complete histogram would require $2^{28 \times 28} - 1$ parameters for $p_{\text{data}}(\mathbf{x})$.

Question: How many parameters are required in these cases?

$$p_{\text{data}}(\mathbf{x}) = p(x_1) \cdot p(x_2) \cdot \cdots \cdot p(x_m);$$

$$p_{\text{data}}(\mathbf{x}) = p(x_1) \cdot p(x_2|x_1) \cdot \cdots \cdot p(x_m|x_{m-1}).$$

Conditional Models

In practice, we're typically interested in learning conditional models (sampling from conditional distribution $p_{\text{data}}(\mathbf{x}|\mathbf{y})$).

Condition $\mathbf{y}$	Output $\mathbf{x}$	Model
$\emptyset$	image	unconditional image model
class label	image	class-conditional image model
text prompt	image	text-to-image model
image	image	image-to-image model
image	text	image-to-text (image captioning) model
English text	Russian text	sequence-to-sequence model (machine translation)
sound	text	speech-to-text (automatic speech recognition) model
text	sound	text-to-speech model

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Divergences

- Let us fix a probabilistic model $p_{\boldsymbol{\theta}}(\mathbf{x})$ from a parametric family of distributions $\{p(\mathbf{x}|\boldsymbol{\theta}) \mid \boldsymbol{\theta} \in \boldsymbol{\Theta}\}$.
- Instead of searching among all possible distributions for the true $p_{\text{data}}(\mathbf{x})$, we seek a functional approximation $p_{\boldsymbol{\theta}}(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$.

What is a Divergence?

Let $\mathcal{P}$ be the set of all probability distributions. A mapping $D : \mathcal{P} \times \mathcal{P} \to \mathbb{R}$ is called a **divergence** if

- $D(\pi \| p) \geq 0$ for all $\pi, p \in \mathcal{P}$
- $D(\pi \| p) = 0$ if and only if $\pi \equiv p$

Divergence Minimization Problem

$$\min_{\boldsymbol{\theta}} D(p_{\text{data}} \| p_{\boldsymbol{\theta}})$$

where $p_{\text{data}}(\mathbf{x})$ is the true data distribution and $p_{\boldsymbol{\theta}}(\mathbf{x})$ is the model distribution.

Forward KL vs Reverse KL (Kullback-Leibler Divergence)

Forward KL

$$D_{\mathrm{KL}}(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = \int p_{\text{data}}(\mathbf{x}) \log \frac{p_{\text{data}}(\mathbf{x})}{p_{\boldsymbol{\theta}}(\mathbf{x})} d\mathbf{x} \to \min_{\boldsymbol{\theta}}$$

Reverse KL

$$D_{\mathrm{KL}}(p_{\boldsymbol{\theta}} \| p_{\text{data}}) = \int p_{\boldsymbol{\theta}}(\mathbf{x}) \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x})} d\mathbf{x} \to \min_{\boldsymbol{\theta}}$$

What's the practical distinction between these two objectives?

Maximum Likelihood Estimation (MLE)

Let $\{\mathbf{x}_i\}_{i=1}^n$ be i.i.d. observed samples.

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \prod_{i=1}^{n} p_{\boldsymbol{\theta}}(\mathbf{x}_i) = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{n} \log p_{\boldsymbol{\theta}}(\mathbf{x}_i).$$

Forward KL vs Reverse KL: MLE as Forward KL

Forward KL

$$
\begin{aligned}
D_{\mathrm{KL}}(p_{\text{data}} \| p_{\boldsymbol{\theta}}) &= \int p_{\text{data}}(\mathbf{x}) \log \frac{p_{\text{data}}(\mathbf{x})}{p_{\boldsymbol{\theta}}(\mathbf{x})} d\mathbf{x} \\
&= \int p_{\text{data}}(\mathbf{x}) \log p_{\text{data}}(\mathbf{x}) d\mathbf{x} - \int p_{\text{data}}(\mathbf{x}) \log p_{\boldsymbol{\theta}}(\mathbf{x}) d\mathbf{x} \\
&= -\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[\log p_{\boldsymbol{\theta}}(\mathbf{x})] + \text{const} \\
&\approx -\frac{1}{n} \sum_{i=1}^{n} \log p_{\boldsymbol{\theta}}(\mathbf{x}_i) + \text{const} \to \min_{\boldsymbol{\theta}}.
\end{aligned}
$$

> **Maximum likelihood estimation** is thus equivalent to minimizing a Monte Carlo estimate of the **forward KL divergence**.

Reverse KL

$$
\begin{aligned}
D_{\mathrm{KL}}(p_{\boldsymbol{\theta}} \| p_{\text{data}}) &= \int p_{\boldsymbol{\theta}}(\mathbf{x}) \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x})} d\mathbf{x} \\
&= \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \left[\log p_{\boldsymbol{\theta}}(\mathbf{x}) - \log p_{\text{data}}(\mathbf{x})\right] \to \min_{\boldsymbol{\theta}}
\end{aligned}
$$

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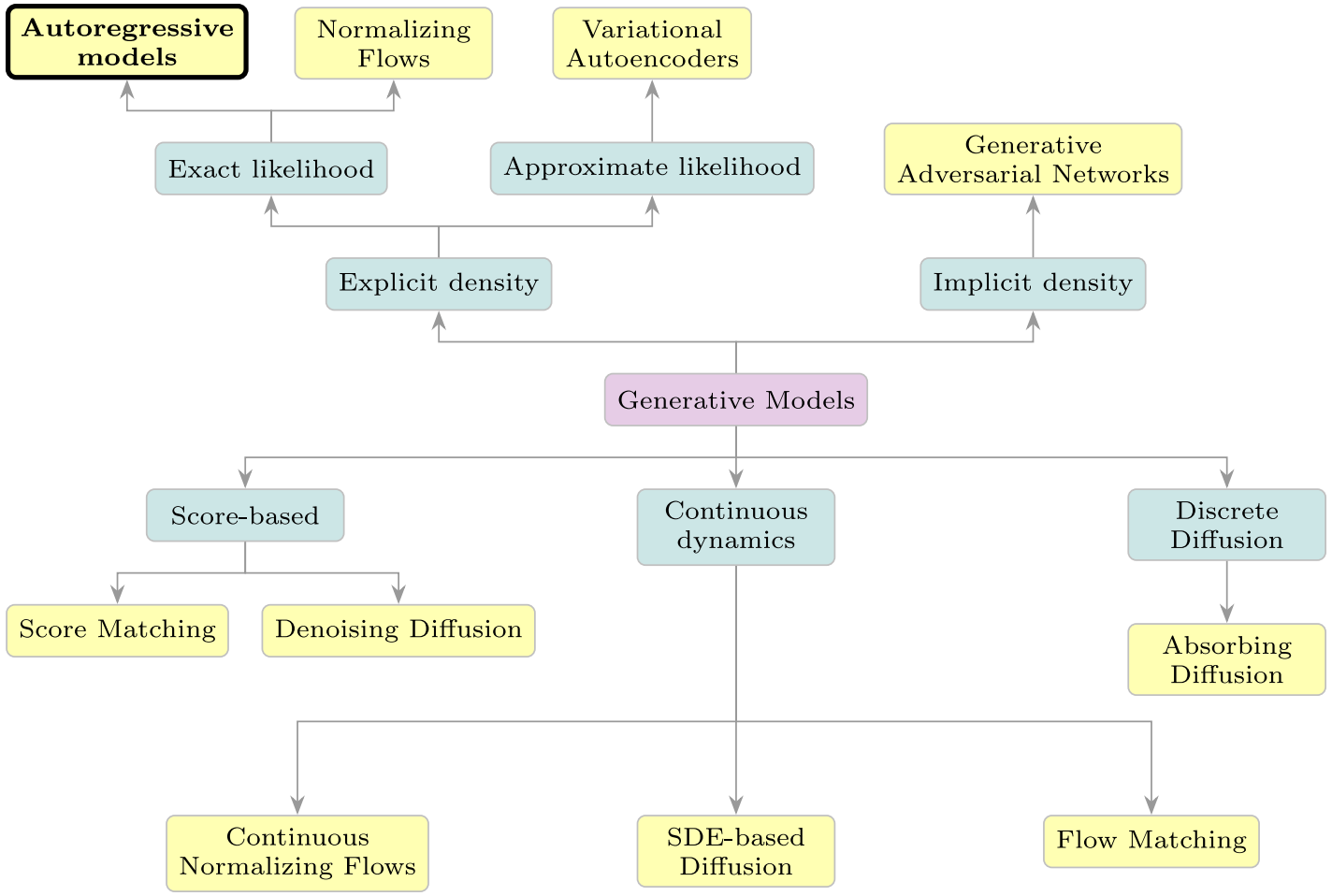


Autoregressive Modeling

MLE Problem

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \prod_{i=1}^{n} p_{\boldsymbol{\theta}}(\mathbf{x}_i) = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{n} \log p_{\boldsymbol{\theta}}(\mathbf{x}_i)$$

- This maximization is typically solved via gradient-based optimization.
- Thus, efficient computation of both $\log p_{\boldsymbol{\theta}}(\mathbf{x})$ and its gradient $\frac{\partial \log p_{\boldsymbol{\theta}}(\mathbf{x})}{\partial \boldsymbol{\theta}}$ is crucial.

Likelihood as a Product of Conditionals

For $\mathbf{x} = (x_1, \ldots, x_m)$, $\mathbf{x}_{1:j} = (x_1, \ldots, x_j)$,

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \prod_{j=1}^{m} p_{\boldsymbol{\theta}}(x_j | \mathbf{x}_{1:j-1}); \quad \log p_{\boldsymbol{\theta}}(\mathbf{x}) = \sum_{j=1}^{m} \log p_{\boldsymbol{\theta}}(x_j | \mathbf{x}_{1:j-1})$$

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{n} \left[\sum_{j=1}^{m} \log p_{\boldsymbol{\theta}}(x_{ij} | \mathbf{x}_{i,1:j-1}) \right]$$

Autoregressive Models

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \sum_{j=1}^{m} \log p_{\boldsymbol{\theta}}(x_j | \mathbf{x}_{1:j-1})$$

Sampling (Ancestral)

1. Sample $\hat{x}_1 \sim p_{\boldsymbol{\theta}}(x_1)$, $\hat{x}_2 \sim p_{\boldsymbol{\theta}}(x_2 | \hat{x}_1)$, ..., $\hat{x}_m \sim p_{\boldsymbol{\theta}}(x_m | \hat{\mathbf{x}}_{1:m-1})$.
2. Return $\hat{\mathbf{x}} = (\hat{x}_1, \hat{x}_2, \ldots, \hat{x}_m)$.

- Each conditional $p_{\boldsymbol{\theta}}(x_j | \mathbf{x}_{1:j-1})$ can be modeled using a neural network.
- Modeling all conditionals separately isn't feasible. To address this, we share parameters across all conditionals.

Sampling vs Density Evaluation

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \prod_{j=1}^{m} p_{\boldsymbol{\theta}}(x_j \mid \mathbf{x}_{1:j-1}), \qquad \log p_{\boldsymbol{\theta}}(\mathbf{x}) = \sum_{j=1}^{m} \log p_{\boldsymbol{\theta}}(x_j \mid \mathbf{x}_{1:j-1}).$$

Raster order · white = 0 · dark = 1
Gray = not sampled

One model, two operations

Sampling: sample a pixel from its conditional distribution; append it to the context.

Density evaluation: use the observed prefix; multiply the probabilities of the observed pixels.

pixel 1	pixel 2	pixel 3	pixel 4	pixel 5	pixel 6	pixel 7	pixel 8	pixel 9
0.563	**0.722**	**1.000**	**0.462**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**

log p(image) = -1.6740

p(image) = 0.18750

> Sampling builds the image pixel by pixel. Density evaluation uses an image that is already known.

Toy data: 32 binary 3 × 3 images with stripes. Conditional probabilities come from counts.

Autoregressive Models: MLP

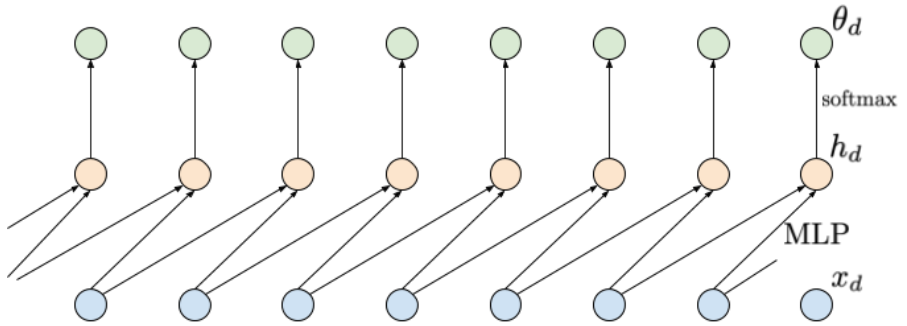
For large j, the conditional $p_{\boldsymbol{\theta}}(x_j|\mathbf{x}_{1:j-1})$ becomes intractable as the history $\mathbf{x}_{1:j-1}$ grows variable-length.

Markov Assumption

$$p_{\boldsymbol{\theta}}(x_j|\mathbf{x}_{1:j-1}) = p_{\boldsymbol{\theta}}(x_j|\mathbf{x}_{j-d:j-1}), \quad d \text{ is a fixed parameter.}$$

Example

- $d = 2$
- $x_j \in \{0, \dots, 255\}$
- $\mathbf{h}_j = \text{MLP}_{\boldsymbol{\theta}}(x_{j-1}, x_{j-2})$
- $\boldsymbol{\pi}_j = \text{softmax}(\mathbf{h}_j)$
- $p_{\boldsymbol{\theta}}(x_j|x_{j-1}, x_{j-2}) = \text{Cat}(\boldsymbol{\pi}_j)$



Can we also model continuous-valued data, not just the discrete case?

Image credit: https://jmtomczak.github.io/blog/2/2_ARM.html

Autoregressive Models: LLM

$$p_{\boldsymbol{\theta}}(x_j|\mathbf{x}_{1:j-1}) = p_{\boldsymbol{\theta}}(x_j|\mathbf{x}_{j-d:j-1}), \quad d \text{ is the context window.}$$

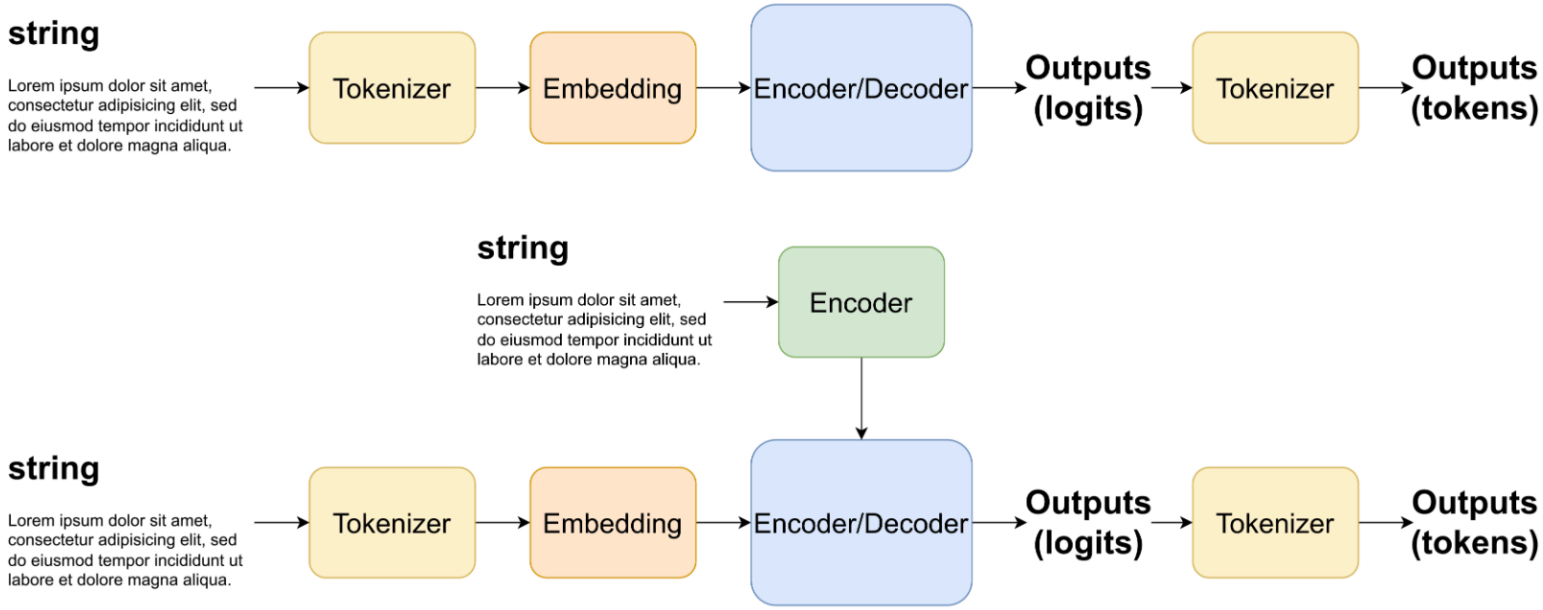


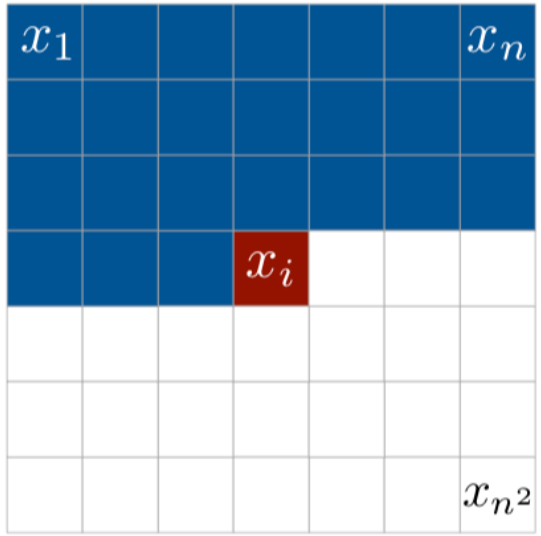
Image credit: https://jmtomczak.github.io/blog/20/20_llms.html

Autoregressive Models for Images

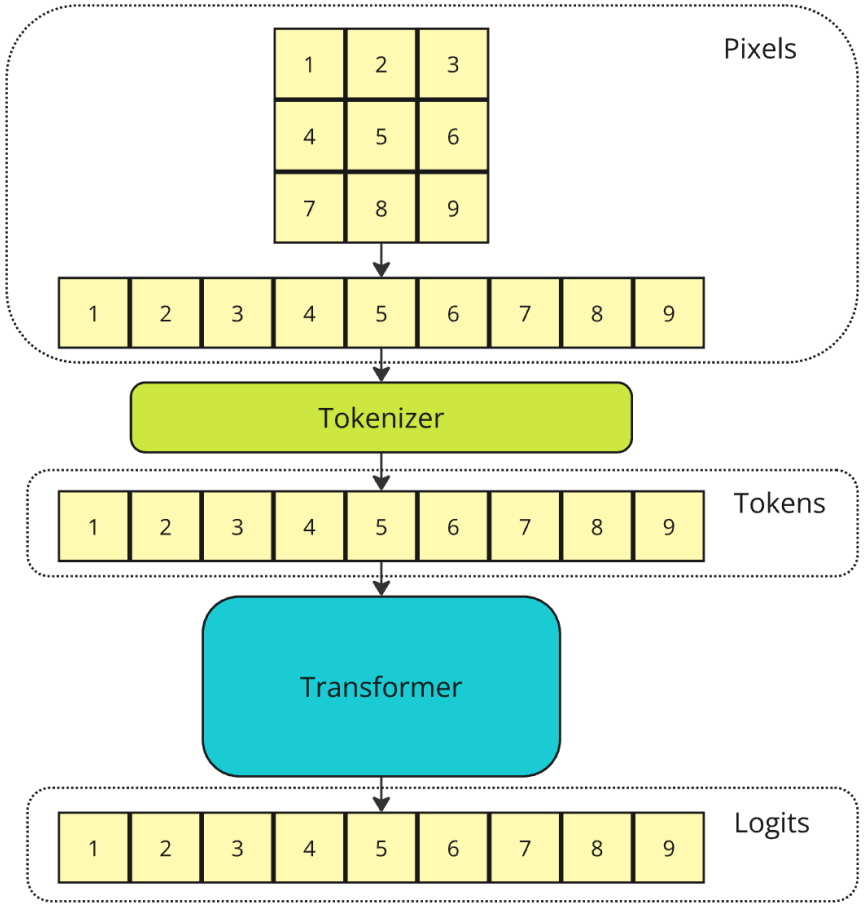
How do we model the distribution $p_{\text{data}}(\mathbf{x})$ of natural images?

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \prod_{j=1}^{\text{width}\times\text{height}} p_{\boldsymbol{\theta}}(x_j | \mathbf{x}_{1:j-1})$$

- A pixel ordering must be selected; the raster scan is a standard choice.
- RGB channel dependencies can be modeled explicitly as well.



Autoregressive Models: ImageGPT



Chen M. et al. Generative Pretraining from Pixels, 2020

Summary

- Density estimation learns a distribution from data; sampling and density evaluation use the fitted model.
- Model families: autoregressive models, normalizing flows, VAEs, GANs, and diffusion models.
- The divergence minimization framework offers a principled way to learn distributions that match the data.
- Minimizing the forward KL divergence is equivalent to maximum likelihood estimation.
- Autoregressive models factorize the joint distribution; density evaluation multiplies the conditionals.
- ImageGPT applies a transformer to raster-ordered image pixels; autoregressive sampling is sequential.